

Mohammed Abdul Omer

Computer Science Engineering (AI and ML)

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PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in artificial intelligence and machine learning, graduating May 2026, with proven experience designing and shipping production-grade generative AI systems, retrieval-augmented generation (RAG) pipelines, multi-agent large language model (LLM) workflows, and end-to-end MLOps automation. Proficient across the complete AI and ML lifecycle from data preprocessing, model training, and hyperparameter tuning through containerized deployment and live performance monitoring. Strong foundation in deep learning, natural language processing (NLP), and computer vision with 6 deployed applications serving real users. Seeking a full-time role as an AI Engineer or ML Engineer.

TECHNICAL SKILLS

Programming Languages: Python, SQL

AI and Machine Learning: supervised learning, unsupervised learning, model training, model evaluation, hyperparameter tuning, feature engineering, data preprocessing, Scikit-learn, PyTorch, TensorFlow

Deep Learning and Computer Vision: convolutional neural networks (CNN), recurrent neural networks (RNN), transfer learning, object detection, image classification, gradient class activation mapping, OpenCV, Hugging Face Transformers

Natural Language Processing (NLP): text classification, information retrieval, large language models (LLM), LangChain, LangGraph, CrewAI, retrieval-augmented generation (RAG), vector databases, prompt engineering, Mistral 7B, LLaMA 3, Groq, Ollama

LLM Fine-Tuning: quantized low-rank adaptation (QLoRA), parameter-efficient fine-tuning (PEFT), supervised fine-tuning, Weights and Biases

Backend and APIs: FastAPI, Django, REST API design, asynchronous programming, Streamlit, Gradio

MLOps and DevOps: MLflow, Weights and Biases, Evidently AI, Docker, model deployment, data drift monitoring

Tools and Platforms: Git, GitHub, Hugging Face Hub, Kaggle, Vercel, Jupyter Notebook, VS Code

Soft Skills: problem solving, analytical thinking, cross-functional collaboration, technical communication, project ownership, adaptability

PROFESSIONAL EXPERIENCE

AI Engineer Intern

March 2025 to June 2025

TechZone Software Academy — Hyderabad, India

- Architected and delivered 4 production-grade AI and ML systems for a live educational platform, streamlining academic workflows and reducing manual overhead across 3 departments
- Designed a retrieval-augmented generation (RAG) tutoring assistant using LangChain, a vector database, and LLaMA 3 via Groq API, applying NLP and information retrieval techniques to significantly cut educator query response time
- Engineered an automated language proficiency test engine with LLM-based scoring across reading, writing, and listening modules, improving student test-readiness through personalised AI feedback
- Built an automated MCQ generation and attendance tracking system that eliminated manual administrative workflows; orchestrated all 4 systems as production-ready FastAPI REST services with continuous reliability

AI and ML Intern

November 2024

RAM Innovative Infotech — Hyderabad, India

- Developed and evaluated a supervised machine learning disease-prediction classifier using Scikit-learn on a structured medical dataset, applying data preprocessing and feature engineering to optimise model accuracy
- Integrated the trained model into a live Django web application, gaining end-to-end experience with REST-based AI model deployment serving real users

PROJECTS

Building Safety Smoke Detection | CNN, object detection, transfer learning, machine learning, Django B.E. Major Project

- Engineered a dual-module fire and smoke detection system on an IoT sensor dataset of 62,630 readings across 13 sensor channels; trained 7 ML classifiers (Random Forest, SVM, Gradient Boosting) achieving AUC-ROC above 0.999
- Applied transfer learning to deploy a MobileNet CNN reaching 96.98% validation accuracy on image classification;

integrated YOLO object detection for real-time bounding box prediction on fire and smoke regions

- Developed and delivered the complete system as a role-based Django web application with Railway deployment

Multi-Document RAG Chatbot | *NLP, LangChain, LangGraph, vector database, FastAPI, Streamlit*

- Implemented a hybrid retrieval agent combining dense semantic embeddings with sparse keyword search and a cross-encoder reranker to improve retrieval precision across multi-document corpora
- Built an asynchronous FastAPI backend with server-sent event streaming for real-time responses; integrated a LangGraph ReAct agent loop with persistent memory for multi-turn conversation; published on Hugging Face Spaces

LLM Fine-Tuning: Mistral 7B Medical QA | *QLoRA, PEFT, supervised fine-tuning, Weights and Biases* *huggingface.co/mohdomer/mistral-7b-medical-qa-qlora*

- Fine-tuned a 7-billion-parameter large language model using 4-bit quantized low-rank adaptation on a Kaggle T4 GPU, cutting GPU memory requirements by over 75% versus full fine-tuning
- Logged all training runs in Weights and Biases; achieved consistent ROUGE-1, ROUGE-2, and BLEU score gains over the untuned baseline on medical question answering benchmarks; published the merged model to Hugging Face Hub

Multi-Agent Research System | *multi-agent AI, CrewAI, Tavily search, LLaMA 3, Python* *huggingface.co/spaces/mohdomer/newresearch*

- Orchestrated a CrewAI multi-agent pipeline with live Tavily web search, coordinating specialised agents for research planning, information retrieval, and structured synthesis; automated the full research workflow from query to formatted report with citations

AI Orchestrator: Multi-Provider LLM Router | *FastAPI, Groq, Gemini, OpenRouter, Ollama* *ai-orchestrator-nu86.onrender.com*

- Designed a unified LLM routing system supporting 17 or more models across 4 providers with ML-based automatic task classification and a benchmark mode for side-by-side latency and output quality comparison

MLOps Pipeline | *MLflow, Weights and Biases, Evidently AI, Docker*

- Constructed a modular, reproducible ML pipeline with MLflow experiment tracking, a model registry for version control, and Evidently AI automated data drift monitoring; containerised the entire workflow with Docker for consistent deployment across environments

PulmoScan AI: Tuberculosis Detection | *CNN, transfer learning, gradient visualization, FastAPI* *pulmoscanai-app.hf.space*

- Trained a tuberculosis detection CNN with transfer learning and gradient-based saliency maps for clinical explainability; automated PDF diagnostic report generation per scan; published as a live application on Hugging Face Spaces

TruthLens: Fake News Detector | *NLP, text classification, CNN-RNN, Keras* *Mini Project — truthlens-uqo4.onrender.com*

- Built a misinformation classifier using a combined CNN and RNN architecture trained on over 40,000 labelled news articles, achieving 94.2% classification accuracy; released as a live inference web application on Render

EDUCATION

Bachelor of Engineering, Computer Science Engineering (AI and ML)

2022 – 2026

Lords Institute of Engineering and Technology, Hyderabad, India

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures and Algorithms, Database Management Systems, Artificial Intelligence, Software Engineering

ACHIEVEMENTS

- Published 5+ deployed AI applications on Hugging Face Spaces and Render with functional inference pipelines accessible to real users
- Fine-tuned and released a Mistral 7B Medical QA model using QLoRA and PEFT with documented ROUGE and BLEU improvements over the untuned baseline
- Built AI systems spanning four domains: Retrieval-Augmented Generation, Computer Vision, Multi-Agent AI, and MLOps end-to-end pipelines
- Delivered production AI solutions supporting academic operations across multiple departments of a live educational platform during internship